VRC-7: Voice Recognition Control for Autonomous Robots

SOTA Compilation, Milestone Plan & Detailed Objectives

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Autonomous Intelligent Systems | Winter 2025/26 | Frankfurt University of Applied Sciences

Supervised by: Prof. Dr. Peter Nauth | Project Period: 01 Dec 2025 – 20 Mar 2026

1. State of the Art (SOTA) Compilation

Research Question: What are the current approaches to voice-controlled robot navigation, and what gaps does VRC-7 address?

1.1 Voice Control in Service Robotics

Early voice-controlled service robots (2000s–2010s) used rigid template-based speech recognition with fixed vocabulary sets. The Nursebot project (Pineau et al., 2003) demonstrated voice-guided navigation in nursing home environments using keyword matching, achieving acceptable accuracy only in quiet, single-accent conditions. HelpMate (Transition Research) and TUG (Aethon) — widely deployed hospital delivery robots — relied on button-press interfaces, implicitly acknowledging the fragility of voice control for diverse user populations.

The fundamental limitation of keyword-based NLU is its inability to handle: (1) accent variation, (2) natural language paraphrasing, and (3) noisy environments. These limitations remain largely unaddressed in deployed healthcare robot systems as of 2024.

1.2 Large Language Model Integration with Robotics

Ahn et al. (2022) introduced SayCan, demonstrating that large language models (LLMs) can ground natural language instructions to robot actions with significantly higher generalisation than keyword matching. SayCan invokes the LLM for every command, producing excellent semantic flexibility. However, this full-cloud approach is impractical under commercial free-tier API rate limits (~30 requests/minute on Groq) for continuous robot operation — quota exhaustion occurs within 4 minutes at typical command frequencies.

Perera et al. (2024) integrated GPT-4 with a socially assistive robot in a geriatric care unit, achieving improved interaction fluency but reporting 2–4 second command latency and prohibitive cost for sustained operation. This motivates the selective LLM invocation design of VRC-7.

1.3 ROS2-Based Voice Control Frameworks

Chatzilygeroudis et al. (2024) presented a voice-enabled ROS2 framework for collaborative inspection using Vosk ASR and a rule-based intent classifier. Their system achieves >90% accuracy in controlled laboratory conditions but does not evaluate accent robustness or ambient noise conditions. The single-layer NLU architecture is functionally equivalent to VRC-7's Layer 1 alone, which achieves only 42% NLU accuracy on accented speech — validating the need for a second layer.

Dominguez-Vidal and Sanfeliu (2024) proposed a ROS-based voice command system for collaborative transportation. Their single-layer approach reports >40% failure rates for non-native English speakers, consistent with VRC-7's baseline measurements. Neither system addresses API efficiency for sustained free-tier operation.

1.4 Accent-Robust Automatic Speech Recognition

Radford et al. (2023) introduced Whisper, a weakly supervised encoder-decoder transformer trained on 680,000 hours of multilingual audio. Whisper demonstrates substantially improved generalisation to accented speech compared to prior ASR systems. However, without language forcing, Whisper's strong multilingual prior causes systematic failure for accented English: the model interprets accented Indian/South Asian English as Icelandic, Danish, or Welsh, producing phonetically plausible but semantically unusable transcriptions (e.g., "turn right" → "Törn rät").

VRC-7 addresses this through two mechanisms: (1) language='en' parameter forces English-only beam search decoding; (2) a vocabulary prompt biases the ASR prior toward robotics-relevant command tokens. Together, these raise transcription accuracy from ~23% to 78% for accented speech — before any NLU recovery.

1.5 Gap Analysis: What VRC-7 Addresses

System	LLM NLU	Accent Robust	API Efficient	Nav. Routing	Key Limitation
Nursebot (2003)	✗	✗	N/A	Fixed	Rigid vocabulary, single accent
SayCan (2022)	✓	Partial	Low	N/A	Every command calls LLM — API quota
Dominguez (2024)	✗	✗	N/A	Fixed	>40% failure for non-native speakers
Chatzilygeroudis (2024)	✗	✗	High	Fixed	No accent/noise evaluation
Perera (2024)	✓	Partial	Low	Fixed	2–4s latency, cost-prohibitive
VRC-7 (this work)	✓	✓	**High**	**Dynamic**	**First to combine selective LLM + vocab-primed ASR + odometry routing**

Novelty claim: To the best of our knowledge, VRC-7 is the first system to combine (1) selective LLM invocation as an accent-recovery layer, (2) vocabulary-primed language-forced ASR, and (3) odometry-based room-exit routing — all within a single open-source ROS2 package targeting autonomous robot delivery.

2. Detailed Project Objectives

Project title: Voice Recognition Control for Autonomous Robot Navigation in Simulation

Core hypothesis: *A voice control system combining accent-robust ASR configuration, dual-layer NLU (local regex + LLM), and structured odometry-based navigation will achieve higher reliability, flexibility, and API efficiency in simulation-based testing compared to standard single-layer ASR/NLU approaches.*

Objective 1: Implement a Robust Voice Control ROS2 Node (U1 – Baseline Accuracy)

Develop a ROS2 Python node (voice_control.py) implementing the complete AI pipeline from microphone input to robot action. The node must:

- Capture audio continuously via callback-based sounddevice.InputStream at 16 kHz without blocking the ROS2 spin thread.
- Apply a two-stage VAD cascade: energy gate (RMS > 0.02) eliminates silence; Silero VAD (p_speech > 0.5) eliminates non-speech audio. Together, these reduce unnecessary ASR API calls by ~60%.
- Transcribe speech using Groq Whisper large-v3-turbo with language='en' and a vocabulary prompt, achieving ≥96% transcription accuracy in quiet conditions (U1).
- Achieve ≥94% end-to-end command execution accuracy for standard English commands in quiet conditions (U1 baseline).
- **Success metric (U1):** ≥94% end-to-end accuracy across 50 standard commands in quiet environment.

Objective 2: Dual-Layer NLU with Accent Robustness (U2 – Accent Robustness)

Implement a dual-layer NLU architecture that achieves high accuracy on accented and non-native English speech without requiring ASR model fine-tuning or retraining:

- **Layer 1 (Local Regex NLU):** 30+ compiled regular expressions covering navigation, movement, turns, cardinal directions, and compound commands. Handles ~85% of commands in <1 ms with zero API cost. Includes accent-variant patterns (e.g., "farmasi" → pharmacy, "take lift" → turn left, "donor" → turn around).
- **Layer 2 (Groq LLaMA 3.3 70B):** Invoked only when Layer 1 returns unknown (~15% of commands). Interprets semantic intent from phonetically garbled or accented transcriptions. Uses temperature=0.0 for deterministic output and structured JSON schema for safe command dispatch.
- **Ablation target:** Language forcing alone raises NLU accuracy from ~20% to 42%; vocabulary priming raises it to ~55%; Layer 2 LLM raises it to 94%. All three mitigations are individually necessary.
- **Success metric (U2):** ≥90% NLU accuracy on accented speech (Indian English, 2+ speakers), measured over 50 trials.

Objective 3: Noise-Resilient Operation (U3 – Noise Robustness)

The system must maintain acceptable command recognition accuracy in environments with ambient background noise (~55 dB), representative of typical lab or office conditions:

- 3× gain normalisation followed by hard clipping to [-1.0, +1.0] compensates for low-gain laptop microphones in noisy conditions, preventing silent-chunk false negatives at the energy VAD gate.
- Noise-word filter discards filler utterances ("okay", "thanks", "hmm") before NLU dispatch, preventing spurious command triggers from acknowledgement speech.
- **Success metric (U3):** ≥76% end-to-end accuracy at 55 dB ambient noise, measured over 50 trials with recorded office/lab background noise.

Objective 4: Vocabulary Scalability Without Code Changes (U4 – Scalability)

The system must support addition of new commands post-deployment without modifying the robot controller code or retraining any model:

- Layer 1 patterns are stored as a Python list — new regex patterns can be appended without touching navigation logic.
- Layer 2 (LLM) inherently interprets semantically reasonable commands through natural language understanding, even without explicit pattern registration.

- Tested by introducing 5 new commands post-evaluation and measuring correct execution without code changes.
- **Success metric (U4):** ≥80% correct execution of 5 new commands added post-deployment, averaged over 10 trials each.

Objective 5: Wall-Safe Odometry Navigation (Navigation Objective)

Implement a navigation algorithm that guarantees collision-free zone-to-zone paths from any arbitrary starting position in the structured hospital corridor environment:

- 4-step room-exit routing algorithm decomposes every navigation path into at most 4 axis-aligned waypoints that exclusively pass through corridor gap positions ($x = \pm 6$ m).
- Proportional heading controller with proximity speed reduction ($\alpha = 0.4$ within 1 m of waypoint) prevents overshoot.
- Wall collision recovery: reverse at $0.5 \times v_{max}$ for 0.8 s, retry up to 3 times before skipping waypoint.
- **Success metric:** 100% navigation success from centre position (0,0); ≥80% from inside rooms; ≥60% from mid-corridor positions.

Objective 6: End-to-End Latency (Responsiveness Objective)

The system must respond to voice commands within a latency budget that maintains user trust and operational responsiveness:

- Audio chunk assembly: 1.5 s (deliberate — ensures complete command capture).
- VAD + ASR: 200–400 ms (Groq Whisper) or 300–500 ms (faster-whisper fallback).
- NLU Layer 1: <1 ms; NLU Layer 2: 200–400 ms (Groq LPU hardware).
- ROS2 dispatch: <5 ms.
- **Success metric:** Total command-to-action latency (excluding chunk assembly) ≤500 ms for 95% of commands in standard operation.

3. Milestone Plan

The project is organised into four phases over 16 weeks (01 Dec 2025 – 20 Mar 2026). Each milestone includes a deliverable, success criterion, and responsible team member.

Phase	Weeks	Milestone	Deliverable / Task	Success Criterion	Lead
1	Wk 1–2	M1: Literature Review	Survey ASR models (Whisper, Vosk), NLU tools, ROS2 audio packages. Document 5+ relevant prior systems.	SOTA table completed; gap analysis written	Both
1	Wk 3–4	M2: Architecture Design	Define ROS2 node graph, data flow (mic → ASR → NLU → /cmd_vel), command schema	Architecture diagram + command schema approved	Mithila

			(JSON), hospital world layout.		
2	Wk 5–6	M3: Simulation Environment	Install ROS2 Humble, TurtleBot3 packages, Gazebo Classic 11. Build hospital_vrc.world (4 zones, corridor gaps, boundary walls).	Robot spawns at (0,0), /odom and /scan topics publishing	Romana
2	Wk 7–8	M4: Audio Pipeline	Implement sounddevice callback, energy VAD, Silero VAD, gain normalisation, queue. Fix WSLg PulseAudio routing.	Microphone captures speech, silent frames discarded, queue filling	Mithila
2	Wk 9	M5: ASR Integration	Integrate Groq Whisper with language='en' + vocab prompt. Implement faster-whisper fallback. Test transcription accuracy.	≥90% transcription accuracy on 20 test commands (quiet)	Mithila
2	Wk 10	M6: Dual-Layer NLU	Implement Layer 1 regex patterns (30+). Implement Layer 2 Groq LLaMA 3.3 70B with structured prompt and JSON parsing.	Layer 1 handles "go to pharmacy"; Layer 2 handles "farmasi"	Mithila
2	Wk 11	M7: Navigation + Integration	Implement 4-step routing algorithm, proportional controller, LiDAR wall detection, concurrency flags. Full pipeline integration test.	Robot navigates ICU→Pharmacy→Ward→Reception from voice commands	Mithila
3	Wk 12	M8: U1 Baseline Test	Run 50 standard command trials (quiet). Measure transcription accuracy, NLU accuracy, E2E accuracy, latency.	E2E accuracy ≥94%; latency ≤500 ms	Both
3	Wk 13	M9: U2+U3 Accent/Noise Test	Run 50 trials each: accented speech (Indian English, 2 speakers) and 55 dB ambient noise. Compare vs U1 baseline.	U2 NLU ≥90%; U3 E2E ≥76%	Both
3	Wk 13	M10: U4 Scalability Test	Add 5 new commands post-evaluation without code changes. Run 10 trials each. Document Layer 1 vs Layer 2 resolution.	≥80% execution accuracy for new commands	Mithila
3	Wk 14	M11: Navigation Tests	Run 5 trials per scenario: centre→all zones, inside	100% from centre; ≥80% from room; ≥60% mid-corridor	Both

			room→other zone, mid-corridor→any zone.		
4	Wk 15	M12: Technical Report	Write IEEE-format paper: introduction, related work, architecture, implementation, results, discussion, conclusion.	Complete draft reviewed by both members	Both
4	Wk 15	M13: Presentation	Prepare 12-slide PowerPoint covering problem, contributions, architecture, results, future work.	Slides reviewed; demo video recorded	Romana
4	Wk 16	M14: Final Submission	Push final code to GitHub. Submit report PDF, presentation, demo recordings, README. Close all open issues.	All deliverables in docs/ and assets/; GitHub repo clean	Both

4. Conclusion Summary

The VRC-7 system, implemented using Groq Whisper large-v3-turbo for ASR (with language forcing and vocabulary priming) and a dual-layer NLU (local regex + Groq LLaMA 3.3 70B), achieved the following results against the stated objectives:

Test	Condition	Target	Achieved	Notes
U1	Quiet, standard English	E2E ≥94%	**94% ✓**	NLU 98%; transcription 96%
U2	Accented speech (Indian English)	NLU ≥90%	**94% ✓**	Layer 1 alone: 42%; +52pp from Layer 2
U3	Background noise (~55 dB)	E2E ≥76%	**76% ✓**	NLU 91%; noise bottleneck is transcription
U4	5 new commands, no code changes	Success ≥80%	**88% ✓**	44/50 trials correct; Layer 2 handles ambiguity
Nav.	Centre position, all zones	Success = 100%	**100% ✓**	20/20 trials; 4-step routing verified
API	LLM call efficiency	Calls reduced ≥80%	**85% ✓**	4 min → 26 min free-tier operation

Primary failure mode: Accumulated odometry drift over extended sessions (>15 min) reduces mid-corridor navigation success from 60% to ~30%. SLAM-based localisation (ROS2 Nav2 + AMCL) is identified as the priority future improvement.

Secondary failure mode: Turn-right commands show the largest accent degradation (86% NLU accuracy) because phonetic distortions "torn rat" and "tyk raut" share no lexical overlap with Layer 1 patterns, requiring consistent Layer 2 escalation.

Overall conclusion: The VRC-7 system demonstrates that the dichotomy between accent robustness and API efficiency is not fundamental — a carefully designed hybrid pipeline achieves both simultaneously. The dual-layer NLU architecture, vocabulary-primed ASR, and odometry-based room-exit routing together constitute a reusable open-source framework applicable to any voice-controlled autonomous robot operating under API rate constraints with diverse-accent user populations.

5. References

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